



CONTROL SYSTEMS

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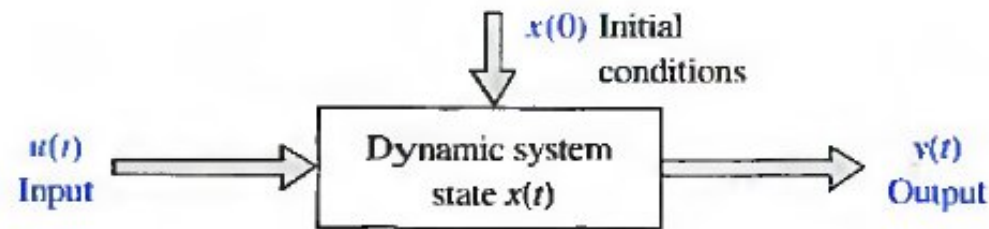
The State Variables of a Dynamic System

Basic Terminologies

Definition:

State variables are the minimum set of independent variables that completely describe the internal condition (state) of a dynamic system at any given time.

Knowing the state variables at time t_0 and the system input for $t \geq t_0$ uniquely determines the future behavior of the system.



The State Variables of a Dynamic System

Basic Terminologies



State of a System

The **state** of a system is the collection of values of the **state variables** at a particular time:

$$\text{State} = \{x_1(t), x_2(t), \dots, x_n(t)\}$$

State Vector

State variables are usually written as a **state vector**:

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

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Basic Terminologies



State Space Model representation

A dynamic system can be modeled using **state equations**:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)$$

Where:

- $\mathbf{x}(t)$ → state vector
- $\mathbf{u}(t)$ → input vector
- $\mathbf{y}(t)$ → output vector
- $\mathbf{A}$ → system matrix
- $\mathbf{B}$ → input matrix
- $\mathbf{C}$ → output matrix
- $\mathbf{D}$ → feedthrough matrix

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Key characteristics of state variables

1. **Minimum number:** Only as many variables as needed to describe the system
2. **Independent:** No state variable can be expressed using others
3. **Energy-related:** Often represent stored energy (electrical, mechanical, thermal)
4. **Define system order:** Number of state variables = **order of the system**

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Examples of State Variables

Electrical Systems

- Inductor current $i_L(t)$
- Capacitor voltage $v_C(t)$

Mechanical Systems

- Position $x(t)$
- Velocity $\dot{x}(t)$

Thermal Systems

- Temperature $T(t)$

The State Variables of a Dynamic System

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Why State Variables are important?

- Allow modeling of **nonlinear** and **time-varying** systems
- Enable **modern control design** (LQR, observer, Kalman filter)
- Directly relate to **physical system behavior**

The State Variables of a Dynamic System

Basic Terminologies



Summary:

State variables are the smallest set of variables whose present values, together with the input, completely determine the future behavior of a dynamic system.

State variables vs transfer function

- State variable approach gives a complete description of the system.
- Transfer function approach gives only the external behavior.
- State variables are suitable for complex, multivariable systems.
- Transfer functions are simpler and useful for basic LTI analysis.
- State-space representation describes the internal dynamics of a system using state variables.
- The transfer function describes only the input-output relationship under zero initial conditions for linear time-invariant systems.

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Basic Terminologies



State variables vs transfer function

Aspect	State Variable Representation	Transfer Function Representation
Basic Idea	Describes system using a set of first-order differential equations	Describes system using ratio of output to input in Laplace domain
Definition	Uses state equations and output equations	Uses a single algebraic equation $G(s) = \frac{Y(s)}{U(s)}$
Mathematical Form	$\dot{x} = Ax + Bu$ $y = Cx + Du$	$G(s) = \frac{\text{Output}}{\text{Input}}$
Initial Conditions	Directly included in state variables	Ignored (assumed zero)
System Type	Works for linear & nonlinear, time-varying & time-invariant systems	Valid only for linear time-invariant (LTI) systems

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The State Variables of a Dynamic System

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State variables vs transfer function

Aspect	State Variable Representation	Transfer Function Representation
Multiple Inputs & Outputs	Easily handles MIMO systems	Mainly suitable for SISO systems
Physical Meaning	State variables often have physical significance	Transfer function has no direct physical meaning
System Order	Order = number of state variables	Order = highest power of s in denominator
Internal Dynamics	Shows internal behavior of the system	Shows only input-output behavior
Control Design	Used in modern control methods (LQR, observers, Kalman filter)	Used in classical control methods (root locus, Bode plot)
Time Domain Analysis	Natural and straightforward	Difficult (requires inverse Laplace)
Frequency Domain Analysis	Possible but indirect	Very convenient
Stability Analysis	Eigenvalues of matrix A	Poles of transfer function

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Basic Terminologies



Example 1: Electrical System (Series RLC Circuit)

Given:

$$R = 2 \Omega, \quad L = 1 H, \quad C = 0.5 F$$

(A) Transfer Function Approach

For output across the capacitor:

$$G(s) = \frac{V_C(s)}{V_{in}(s)}$$
$$G(s) = \frac{1}{LCs^2 + RCs + 1}$$

Substitute values:

$$G(s) = \frac{1}{(1)(0.5)s^2 + (2)(0.5)s + 1}$$

$$G(s) = \frac{1}{0.5s^2 + s + 1}$$

This shows **only input-output behavior** and assumes **zero initial conditions**.

(B) State-Variable Approach

Choose state variables:

- $x_1 = i_L$ (inductor current)
- $x_2 = v_C$ (capacitor voltage)

State equations:

$$\dot{x}_1 = -2x_1 - x_2 + u$$

$$\dot{x}_2 = 2x_1$$

Matrix form:

$$\dot{x} = \begin{bmatrix} -2 & -1 \\ 2 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

Output equation:

$$y = [0 \quad 1]x$$

This shows **internal dynamics** and allows **non-zero initial conditions**.

The State Variables of a Dynamic System

Basic Terminologies



Example 2: Mechanical System (Mass–Spring–Damper)

Given:

$$m = 1 \text{ kg}, \quad c = 4 \text{ Ns/m}, \quad k = 5 \text{ N/m}$$

(A) Transfer Function

Equation of motion:

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

Taking Laplace transform:

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{s^2 + 4s + 5}$$

(B) State-Variable Model

Choose:

- $x_1 = x$ (position)
- $x_2 = \dot{x}$ (velocity)

State equations:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -5x_1 - 4x_2 + u$$

Matrix form:

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -5 & -4 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

Output:

$$y = [1 \quad 0]x$$

The State Variables of a Dynamic System

Basic Terminologies



Key Takeaway from Numerical Examples

Feature	Transfer Function	State Variables
Initial conditions	Not included	Included
Internal states	Hidden	Explicit
MIMO capability	Limited	Natural
Physical meaning	Abstract	Clear

A transfer function gives a compact input–output description under zero initial conditions, while state variables provide a complete time-domain description including internal system dynamics.

STATE VARIABLE MODELS

Networks to State-space Model



The State Variables of a Dynamic System

Basic Terminologies



State Variables for **RLC Circuits**

Basic Rule

The number of state variables in an RLC circuit equals the number of independent energy-storage elements.

Energy-storage elements:

- Inductor (L) → stores magnetic energy
- Capacitor (C) → stores electric energy

Resistors do **not** store energy → **not state variables**.

The State Variables of a Dynamic System

Basic Terminologies



State Variables for RLC Circuits

1. Series RLC Circuit

Circuit:

Input voltage $u(t)$ applied to series R, L, C

Choice of State Variables:

$$x_1 = i_L(t)$$

$$x_2 = v_C(t)$$

State Equations:

Using KVL:

$$u(t) = Ri_L + L \frac{di_L}{dt} + v_C$$

Capacitor relation:

$$i_L = C \frac{dv_C}{dt}$$

Rewriting:

$$\dot{x}_1 = -\frac{R}{L}x_1 - \frac{1}{L}x_2 + \frac{1}{L}u$$

$$\dot{x}_2 = \frac{1}{C}x_1$$

Matrix Form (State Equation)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} u$$

Output Equation (example: output = capacitor voltage)

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The State Variables of a Dynamic System

Basic Terminologies



State Variables for **RLC Circuits**

Matrix Form (State Equation)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{RC} & -\frac{1}{C} \\ \frac{1}{L} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{C} \\ 0 \end{bmatrix} u$$

Output Equation (example: output = capacitor voltage)

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

2. Parallel RLC Circuit

Circuit:

Input current $u(t)$ applied to parallel R, L, C

Choice of State Variables:

$$x_1 = v_C(t)$$

$$x_2 = i_L(t)$$

State Equations:

Using KCL:

$$u(t) = \frac{v_C}{R} + i_L + C \frac{dv_C}{dt}$$

Inductor relation:

$$v_C = L \frac{di_L}{dt}$$

Rewriting:

$$\begin{aligned} \dot{x}_1 &= -\frac{1}{RC}x_1 - \frac{1}{C}x_2 + \frac{1}{C}u \\ \dot{x}_2 &= \frac{1}{L}x_1 \end{aligned}$$

The State Variables of a Dynamic System

Basic Terminologies



State Variables for **RLC Circuits**

3. Order of RLC Circuits

Circuit	Energy Storage Elements	Order	No. of State Variables
RL or RC	1	First order	1
RLC	2	Second order	2

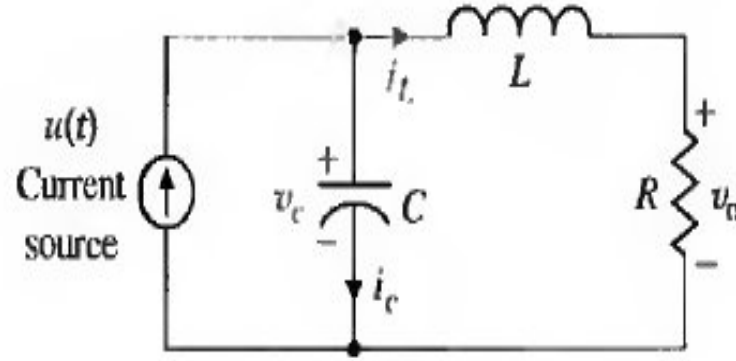
State Differential Equation

State Space Model – Example-1

$$i_c = C \frac{dv_c}{dt} = +u(t) - i_L.$$

$$L \frac{di_L}{dt} = -Ri_L + v_c.$$

$$v_o = Ri_L(t).$$



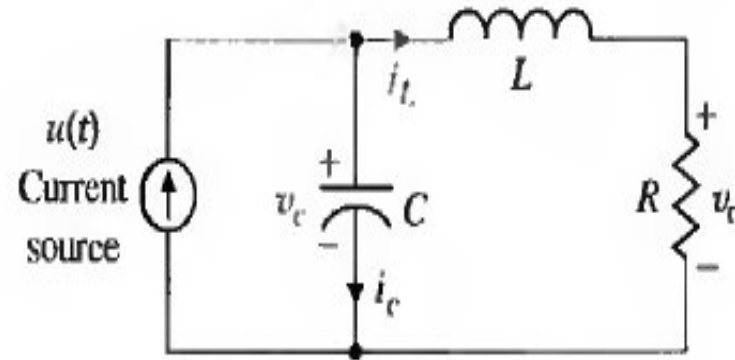
State Differential Equation

State Space Model – Example-1

$$i_c = C \frac{dv_c}{dt} = +u(t) - i_L.$$

$$L \frac{di_L}{dt} = -Ri_L + v_c.$$

$$v_o = Ri_L(t).$$



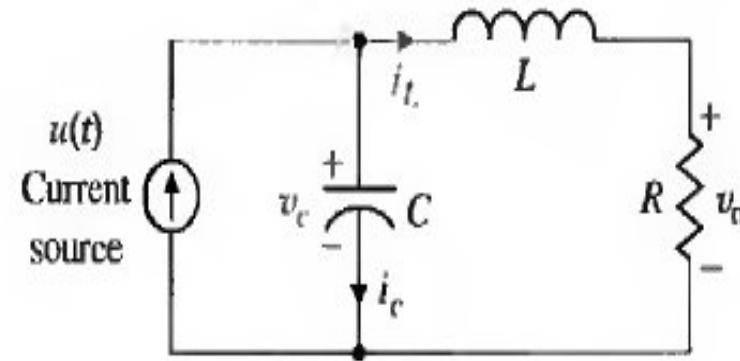
State Differential Equation

State Space Model – Example-1

$$\frac{dx_1}{dt} = -\frac{1}{C}x_2 + \frac{1}{C}u(t),$$

$$\frac{dx_2}{dt} = +\frac{1}{L}x_1 - \frac{R}{L}x_2.$$

$$y_1(t) = v_o(t) = Rx_2.$$



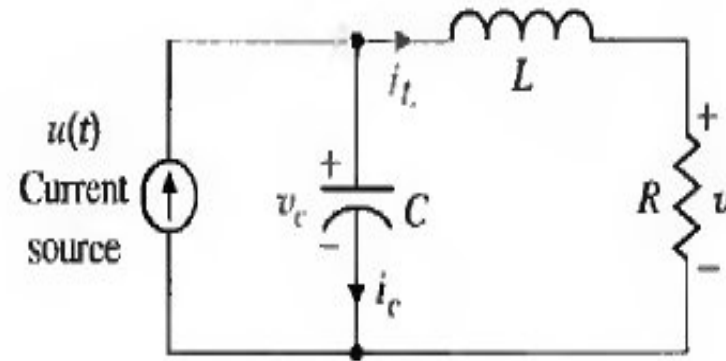
State Differential Equation

State Space Model – Example-1

- In compact form,

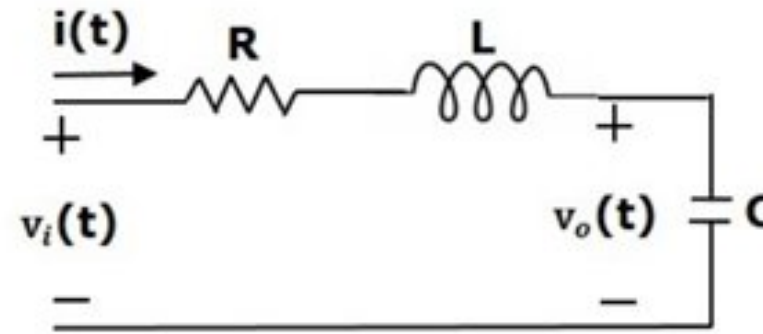
$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & \frac{-1}{C} \\ \frac{1}{L} & \frac{-R}{L} \end{bmatrix} \mathbf{x} + \begin{bmatrix} \frac{1}{C} \\ 0 \end{bmatrix} u(t)$$

$$y = [0 \quad R] \mathbf{x}.$$



State Differential Equation

State Space Model – Example 2



Apply KVL around the loop.

$$v_i(t) = Ri(t) + L \frac{di(t)}{dt} + v_c(t)$$
$$\Rightarrow \frac{di(t)}{dt} = -\frac{Ri(t)}{L} - \frac{v_c(t)}{L} + \frac{v_i(t)}{L}$$

The voltage across the capacitor is -

$$v_c(t) = \frac{1}{C} \int i(t) dt$$

State Differential Equation

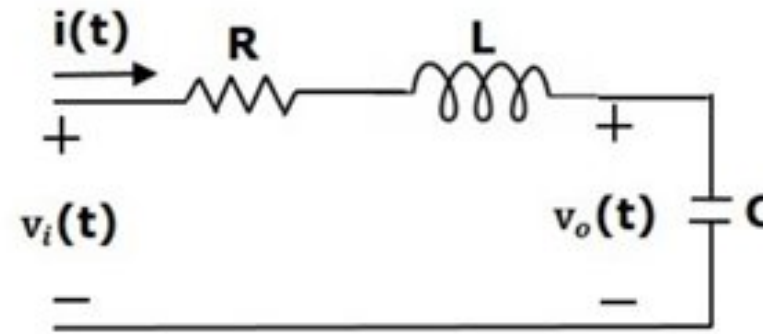
State Space Model – Example 2

Differentiate the above equation with respect to time.

$$\frac{dv_c(t)}{dt} = \frac{i(t)}{C}$$

State vector, $X = \begin{bmatrix} i(t) \\ v_c(t) \end{bmatrix}$

Differential state vector, $\dot{X} = \begin{bmatrix} \frac{di(t)}{dt} \\ \frac{dv_c(t)}{dt} \end{bmatrix}$



State Differential Equation

State Space Model – Example 2

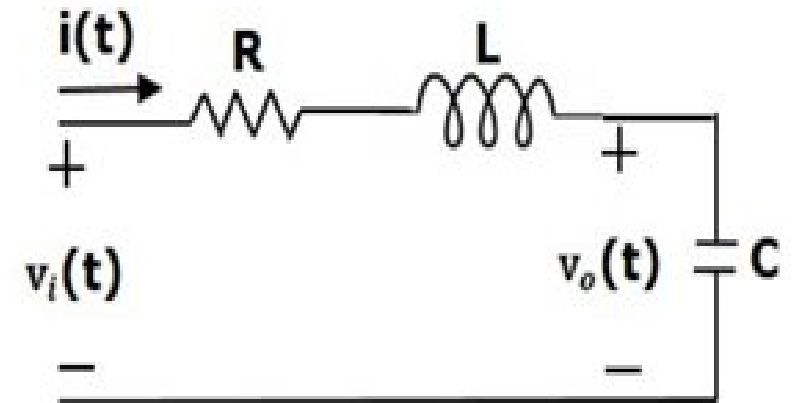


We can arrange the differential equations and output equation into standard form of state space model as,

$$\dot{X} = \begin{bmatrix} \frac{di(t)}{dt} \\ \frac{dv_c(t)}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} i(t) \\ v_c(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} [v_i(t)]$$
$$Y = [0 \quad 1] \begin{bmatrix} i(t) \\ v_c(t) \end{bmatrix}$$

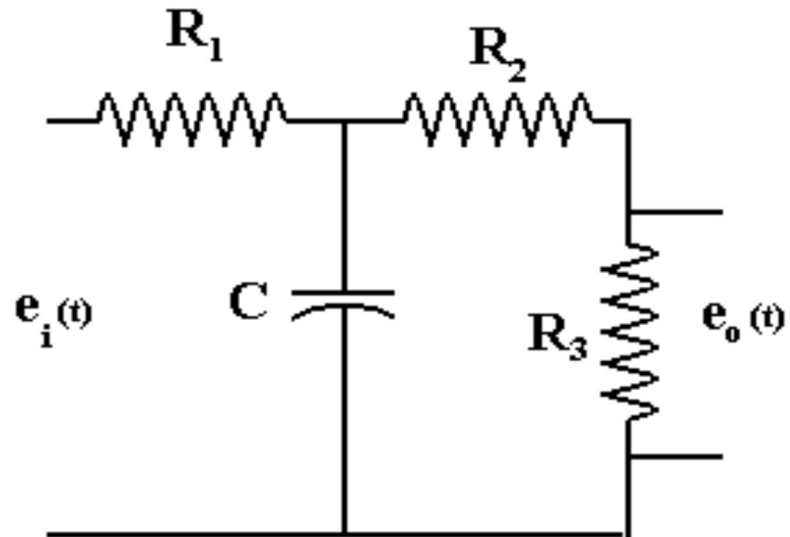
Where,

$$A = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix}, B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}, C = [0 \quad 1] \text{ and } D = [0]$$



State Differential Equation

State Space Model – Example 3



STATE VARIABLE MODELS

Transfer Function to State-space Model



STATE VARIABLE MODELS

Conversion from Transfer Function to State-space Model



$$T(s) = \frac{Y(s)}{U(s)} = \frac{b_3s^3 + b_2s^2 + b_1s + b_0}{s^4 + a_3s^3 + a_2s^2 + a_1s + a_0}$$

Steps:

Number of state variables= order of the denominator polynomial.

Write the differential equation

Multiply both sides by the denominator:

$$(s^4 + a_3s^3 + a_2s^2 + a_1s + a_0)Y(s) = (b_3s^3 + b_2s^2 + b_1s + b_0)U(s)$$

Taking inverse Laplace:

$$y^{(4)} + a_3y^{(3)} + a_2y'' + a_1y' + a_0y = b_3u^{(3)} + b_2u'' + b_1u' + b_0u$$

STATE VARIABLE MODELS

Conversion from Transfer Function to State-space Model



Define state variables

Choose:

$$\begin{aligned}x_1 &= y \\x_2 &= \dot{y} \\x_3 &= \ddot{y} \\x_4 &= y^{(3)}\end{aligned}$$

Then:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= -a_0x_1 - a_1x_2 - a_2x_3 - a_3x_4 + u\end{aligned}$$

STATE VARIABLE MODELS

Conversion from Transfer Function to State-space Model



Output equation

From the numerator:

$$y = b_0x_1 + b_1x_2 + b_2x_3 + b_3x_4$$

State-Space Model

State equation

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 & -a_3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u$$

Output equation

$$y = [b_0 \quad b_1 \quad b_2 \quad b_3] x$$

STATE VARIABLE MODELS

Conversion from Transfer Function to State-space Model

$$T(s) = \frac{12s^3 + 36s - 21}{3s^5 + 9s^3 - 15s^2 + 24s + 18}$$

$$= \frac{4s^3 + 12s - 7}{s^5 + 3s^3 - 5s^2 + 8s + 6}$$

STATE VARIABLE MODELS

Conversion from Transfer Function to State-space Model

State Equation

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -6 & -8 & 5 & -3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u$$

Output Equation

$$y = \begin{bmatrix} -7 & 12 & 0 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

STATE VARIABLE MODELS

Differential equations to State-space Model



STATE VARIABLE MODELS

Conversion from Differential equation to State-space Model



$$4\frac{d^3y(t)}{dt^3} + 2\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + 2y(t) = 5r(t)$$

Choose state variables

Let:

$$x_1 = y$$

$$x_2 = \dot{y}$$

$$x_3 = \ddot{y}$$

Then:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -\frac{1}{2}x_1 - \frac{1}{4}x_2 - \frac{1}{2}x_3 + \frac{5}{4}r(t)$$

STATE VARIABLE MODELS

Conversion from Differential equation to State-space Model



Output equation

$$y = x_1$$

State-Space Model

State equation

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{1}{2} & -\frac{1}{4} & -\frac{1}{2} \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \frac{5}{4} \end{bmatrix} r(t)$$

Output equation

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x$$

STATE VARIABLE MODELS

State-space Model to Differential equations



STATE VARIABLE MODELS

Conversion from State-space Model to differential equation



Choose state variables

$$x_1 = c$$

$$x_2 = \dot{c}$$

$$x_3 = \ddot{c}$$

$$x_4 = \dddot{c}$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = -2x_1 - x_2 - 2x_3 - 4x_4 + 10r(t)$$

STATE VARIABLE MODELS

Conversion from State-space Model to differential equation



Output equation

$$c(t) = x_1$$

State-Space Model

State equation

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & -1 & -2 & -4 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 10 \end{bmatrix} r(t)$$

Output equation

$$c = [1 \ 0 \ 0 \ 0] x$$

STATE VARIABLE MODELS

Conversion from State-space Model to differential equation



Given state-space model

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -3x_1 - 4x_2 + u(t) \\ y(t) &= x_1\end{aligned}$$

Step 1: Express states in terms of output

Since $y = x_1$,

$$\begin{aligned}x_1 &= y \\ x_2 &= \dot{y}\end{aligned}$$

Step 2: Differentiate to eliminate states

$$\dot{x}_2 = \ddot{y}$$

Step 3: Substitute into state equation

From $\dot{x}_2$:

$$\ddot{y} = -3y - 4\dot{y} + u(t)$$

Differential Equation

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = u(t)$$

STATE VARIABLE MODELS

Conversion from State-space Model to differential equation



State equations

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -2x_1 - 3x_2 - x_3 + r(t)$$

$$y = x_1$$

Differential equation

$$\boxed{\frac{d^3y}{dt^3} + \frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = r(t)}$$

STATE VARIABLE MODELS

Conversion from State-space Model to differential equation



Given state-space model

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & -1 & -2 & -4 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 10 \end{bmatrix} r(t), \quad c(t) = x_1$$

Express states in terms of output

$$c(t) = x_1$$

$$x_1 = c$$

$$x_2 = \dot{c}$$

$$x_3 = \ddot{c}$$

$$x_4 = \dddot{c}$$

Write the state equations

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = -2x_1 - x_2 - 2x_3 - 4x_4 + 10r(t)$$

Substitute into the last state equation

$$\dot{x}_4 = \dddot{c}$$

$$\dddot{c} = -2c - \dot{c} - 2\ddot{c} - 4\dddot{c} + 10r(t)$$

$$\ddot{c} + 4\ddot{c} + 2\ddot{c} + \dot{c} + 2c = 10r(t)$$

$$\frac{d^4 c(t)}{dt^4} + 4\frac{d^3 c(t)}{dt^3} + 2\frac{d^2 c(t)}{dt^2} + \frac{dc(t)}{dt} + 2c(t) = 10r(t)$$

STATE VARIABLE MODELS

Conversion from State-space Model to differential equation



State equations

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -2x_1 - 3x_2 - x_3 + r(t)$$

$$y = x_1$$

Differential equation

$$\frac{d^3y}{dt^3} + \frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = r(t)$$

STATE VARIABLE MODELS

Signal Flow Graphs to State-space Model



STATE VARIABLE MODELS

Signal Flow Graph State Models

- First we note that the system is fourth order, and hence we identify four state variables (x_1, x_2, x_3, x_4) .

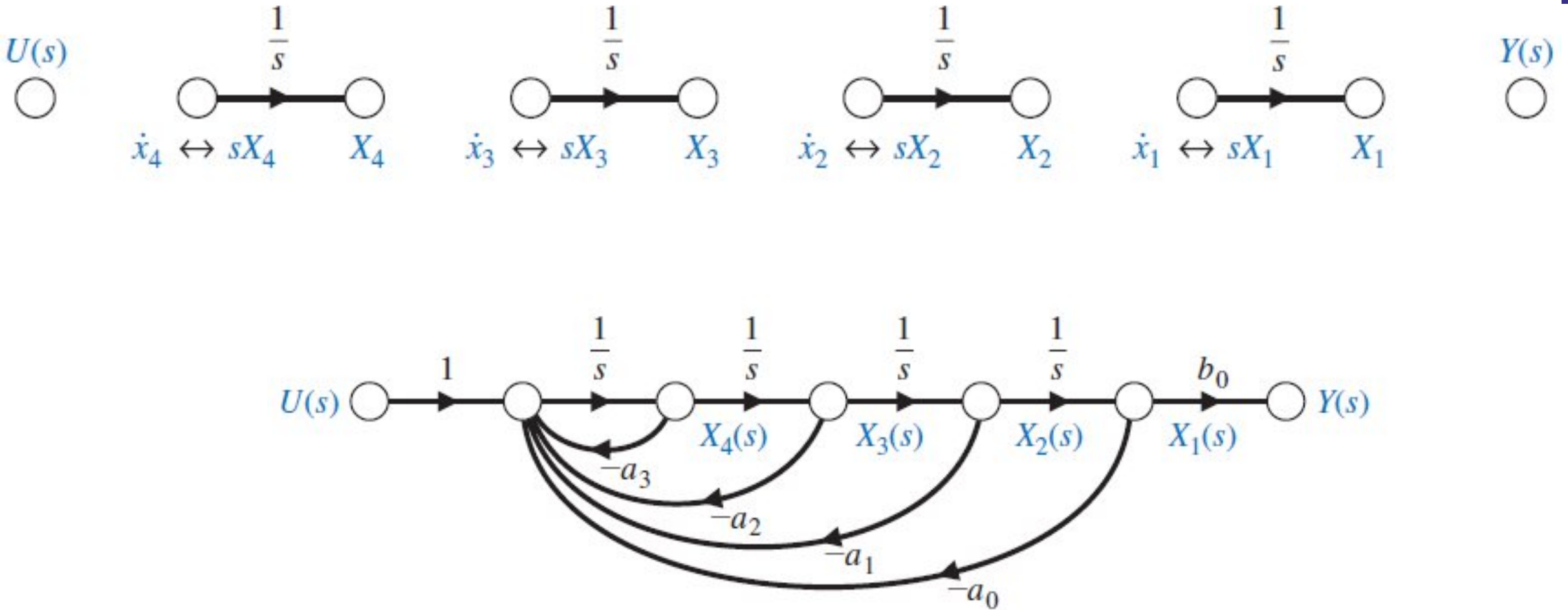
$$\begin{aligned} G(s) &= \frac{Y(s)}{U(s)} = \frac{b_0}{s^4 + a_3s^3 + a_2s^2 + a_1s + a_0} \\ &= \frac{b_0s^{-4}}{1 + a_3s^{-1} + a_2s^{-2} + a_1s^{-3} + a_0s^{-4}}. \end{aligned}$$

When all the feedback loops are touching and all the forward paths touch the feedback loops,

$$G(s) = \frac{\sum_k P_k(s)}{1 - \sum_{q=1}^N L_q(s)} = \frac{\text{Sum of the forward - path factors}}{1 - \text{sum of the feedback loop factors}}.$$

STATE VARIABLE MODELS

Signal Flow Graph State Models



STATE VARIABLE MODELS

Signal Flow Graph State Models

$$\frac{d^4(y(t)/b_0)}{dt^4} + a_3 \frac{d^3(y(t)/b_0)}{dt^3} + a_2 \frac{d^2(y(t)/b_0)}{dt^2} + a_1 \frac{d(y(t)/b_0)}{dt} + a_0(y(t)/b_0) = u(t).$$

Define the four state variables as follows:

$$x_1(t) = y(t)/b_0$$

$$x_2(t) = \dot{x}_1(t) = \dot{y}(t)/b_0$$

$$x_3(t) = \dot{x}_2(t) = \ddot{y}(t)/b_0$$

$$x_4(t) = \dot{x}_3(t) = \dddot{y}(t)/b_0.$$

STATE VARIABLE MODELS

Signal Flow Graph State Models

$$\dot{x}_1(t) = x_2(t),$$

$$\dot{x}_2(t) = x_3(t),$$

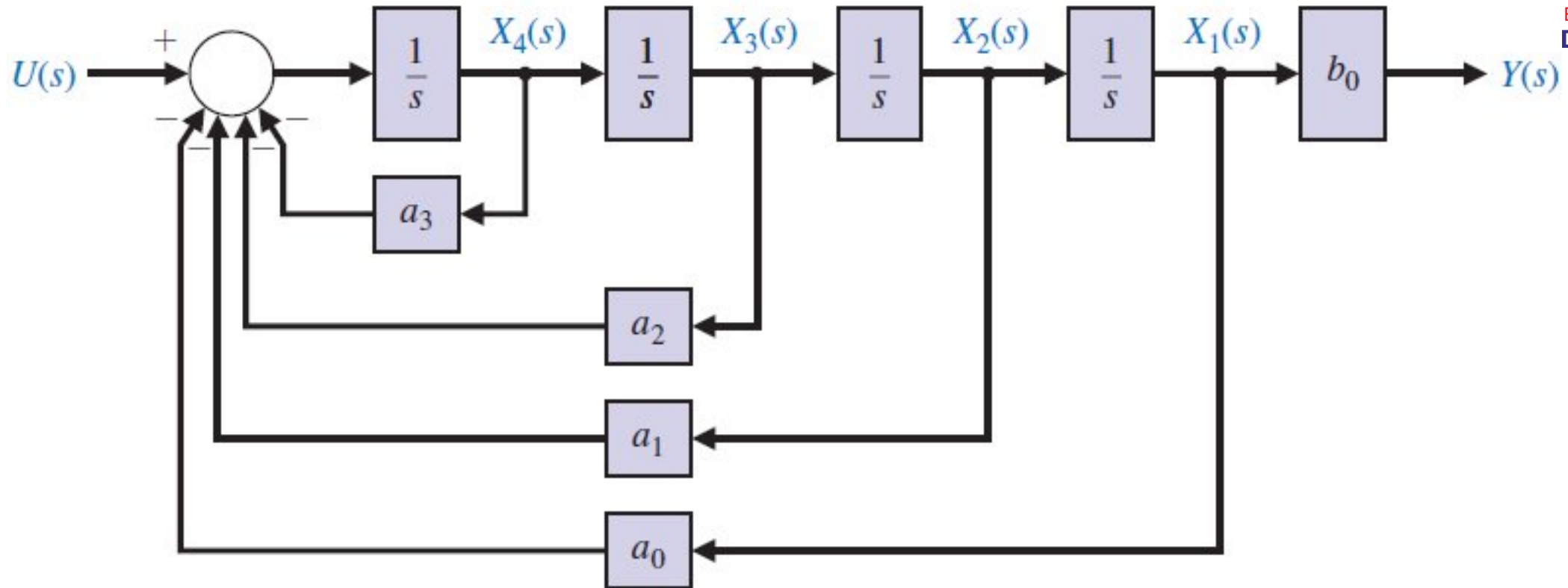
$$\dot{x}_3(t) = x_4(t),$$

$$\dot{x}_4(t) = -a_0x_1(t) - a_1x_2(t) - a_2x_3(t) - a_3x_4(t) + u(t)$$

$$y(t) = b_0x_1(t).$$

STATE VARIABLE MODELS

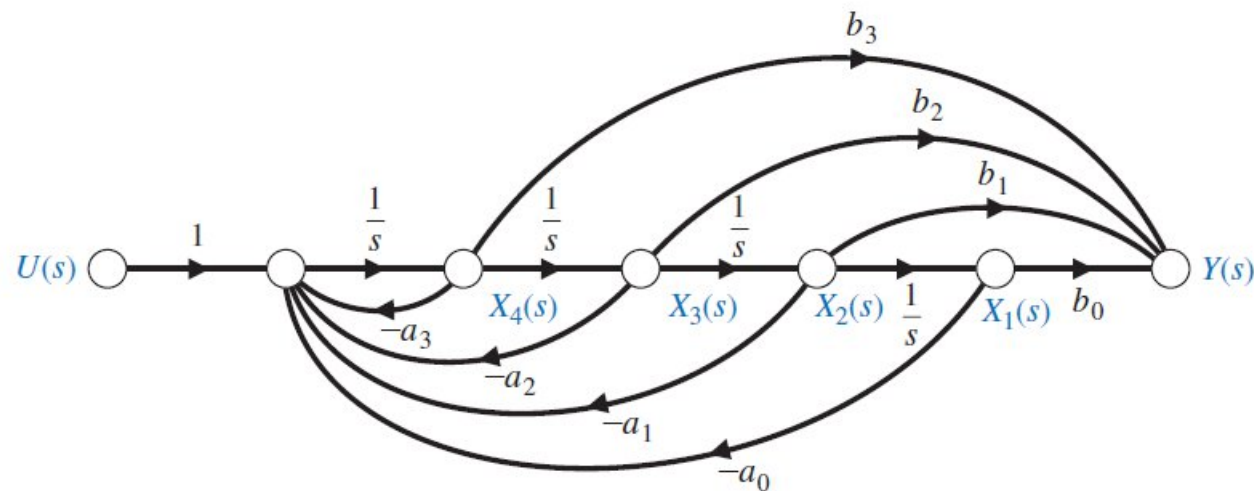
Signal Flow Graph State Models



STATE VARIABLE MODELS

Signal Flow Graph State Models

$$G(s) = \frac{b_3s^3 + b_2s^2 + b_1s + b_0}{s^4 + a_3s^3 + a_2s^2 + a_1s + a_0}$$
$$= \frac{b_3s^{-1} + b_2s^{-2} + b_1s^{-3} + b_0s^{-4}}{1 + a_3s^{-1} + a_2s^{-2} + a_1s^{-3} + a_0s^{-4}}$$



STATE VARIABLE MODELS

The solution of state equation



STATE VARIABLE MODELS

The Time Response and The State Transition Matrix for homogeneous System



1. **Start with the State Equation:**

$$\dot{x}(t) = Ax(t)$$

2. **Apply the Laplace Transform:** Recall that $\mathcal{L}\{\dot{x}(t)\} = sX(s) - x(0)$.

$$sX(s) - x(0) = AX(s)$$

3. **Group the $X(s)$ terms:** Bring all terms involving the state vector to one side. Note that we must use the Identity matrix (I) to keep the dimensions consistent.

$$sIX(s) - AX(s) = x(0)$$

$$(sI - A)X(s) = x(0)$$

STATE VARIABLE MODELS

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4. **Solve for $X(s)$:** Multiply by the inverse of the characteristic matrix $(sI - A)$.

$$X(s) = (sI - A)^{-1}x(0)$$

5. **Apply the Inverse Laplace Transform:** To get back to the time domain, we define the **State Transition Matrix $\phi(t)$** .

$$x(t) = \mathcal{L}^{-1}\{(sI - A)^{-1}\}x(0)$$

Result: $x(t) = \phi(t)x(0)$

STATE VARIABLE MODELS

The Time Response and The State Transition Matrix for homogeneous System



1. **Start with the State Equation:**

$$\dot{x}(t) - Ax(t) = 0$$

2. **Premultiply by the Integrating Factor:** We use e^{-At} to "collapse" the left side of the equation.

$$e^{-At}\dot{x}(t) - e^{-At}Ax(t) = 0$$

3. **Recognize the Product Rule:** The left side is exactly the derivative of the product $e^{-At}x(t)$.

$$\frac{d}{dt}[e^{-At}x(t)] = 0$$

STATE VARIABLE MODELS

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4. Integrate Both Sides

To isolate the state vector, we integrate the expression with respect to time (using a dummy variable τ) from the initial time 0 to the current time t :

$$\int_0^t \frac{d}{d\tau} [e^{-A\tau} x(\tau)] d\tau = 0$$

5. Evaluate the Definite Integral

Applying the Fundamental Theorem of Calculus, we evaluate the expression at the limits of integration:

$$[e^{-A\tau} x(\tau)] \Big|_0^t = 0$$

$$e^{-At} x(t) - e^{-A(0)} x(0) = 0$$

Since $e^0 = I$ (the identity matrix), this simplifies to:

$$e^{-At} x(t) - x(0) = 0$$

STATE VARIABLE MODELS

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6. Solve for $x(t)$

Finally, move $x(0)$ to the right side and premultiply by e^{At} to obtain the general solution:

$$e^{-At}x(t) = x(0)$$

$$\boxed{x(t) = e^{At}x(0)}$$

This result shows that the state at any time t is the initial state $x(0)$ transformed by the **State Transition Matrix**, $\phi(t) = e^{At}$.

Example: If $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, then $e^{At} = \begin{bmatrix} e^{2t} & 0 \\ 0 & e^{3t} \end{bmatrix}$

STATE VARIABLE MODELS

The Time Response and The State Transition Matrix for homogeneous System



State Transition Matrix

$$\phi(t) = e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$

Find the solution of the given state equation:

$$\dot{x} = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} x, \quad x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Method-1

1. Find $(sI - A)$

$$(sI - A) = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} s & 2 \\ -1 & s+3 \end{bmatrix}$$

2. Find the inverse $(sI - A)^{-1}$

The formula for a 2×2 inverse is $\frac{1}{\det(M)} \text{adj}(M)$:

- **Determinant:** $s(s + 3) - (2)(-1) = s^2 + 3s + 2 = (s + 1)(s + 2)$.
- **Adjoint:** $\begin{bmatrix} s + 3 & -2 \\ 1 & s \end{bmatrix}$

$$(sI - A)^{-1} = \frac{1}{(s + 1)(s + 2)} \begin{bmatrix} s + 3 & -2 \\ 1 & s \end{bmatrix} = \begin{bmatrix} \frac{s+3}{(s+1)(s+2)} & \frac{-2}{(s+1)(s+2)} \\ \frac{1}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix}$$

3. Find the State Transition Matrix $\phi(t)$

Apply the Inverse Laplace Transform to each element using partial fraction expansion:

- $\mathcal{L}^{-1} \left[\frac{s+3}{(s+1)(s+2)} \right] = 2e^{-t} - e^{-2t}$
- $\mathcal{L}^{-1} \left[\frac{-2}{(s+1)(s+2)} \right] = -2e^{-t} + 2e^{-2t}$
- $\mathcal{L}^{-1} \left[\frac{1}{(s+1)(s+2)} \right] = e^{-t} - e^{-2t}$
- $\mathcal{L}^{-1} \left[\frac{s}{(s+1)(s+2)} \right] = -e^{-t} + 2e^{-2t}$

$$\phi(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & -2e^{-t} + 2e^{-2t} \\ e^{-t} - e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

4. Find the final solution $x(t)$

Multiply the state transition matrix by the initial condition $x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

$$x(t) = \phi(t) \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

STATE VARIABLE MODELS

The Time Response and The State Transition Matrix



Method-2 to find STM

Step 1: Eigenvalues

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ -2 & -\lambda \end{vmatrix} = \lambda^2 + 2 = 0$$
$$\lambda = \pm i\sqrt{2}$$

Step 2: Use the eigenvalue formula

For complex conjugate eigenvalues $\alpha \pm i\beta$,

$$e^{At} = e^{\alpha t} \left[I \cos(\beta t) + \frac{1}{\beta} (A - \alpha I) \sin(\beta t) \right]$$

Here: $\alpha = 0$, $\beta = \sqrt{2}$

$$e^{At} = I \cos(\sqrt{2}t) + \frac{A}{\sqrt{2}} \sin(\sqrt{2}t)$$

STATE VARIABLE MODELS

The Time Response and The State Transition Matrix



Method-2 to find STM

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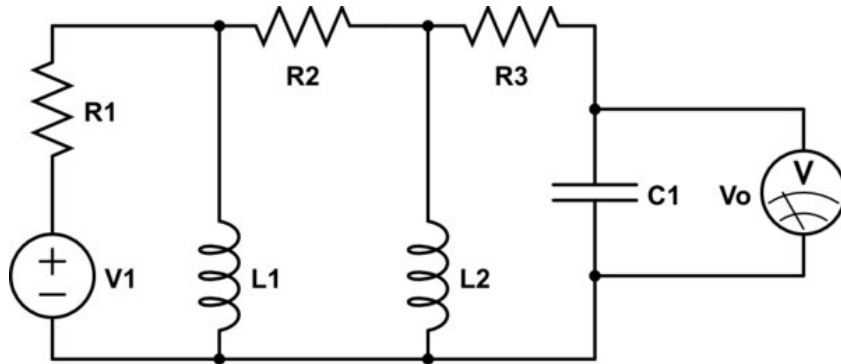
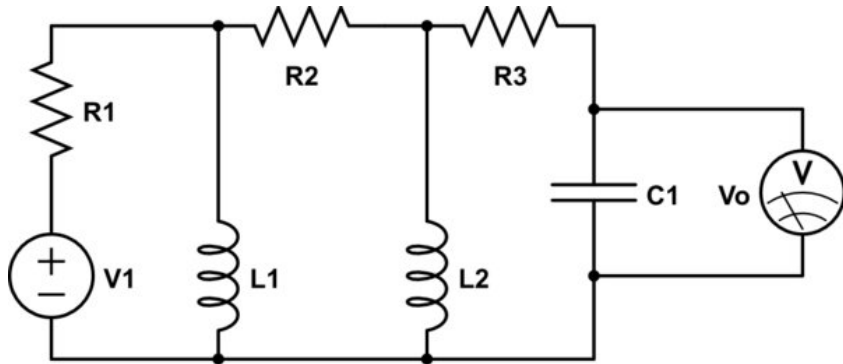
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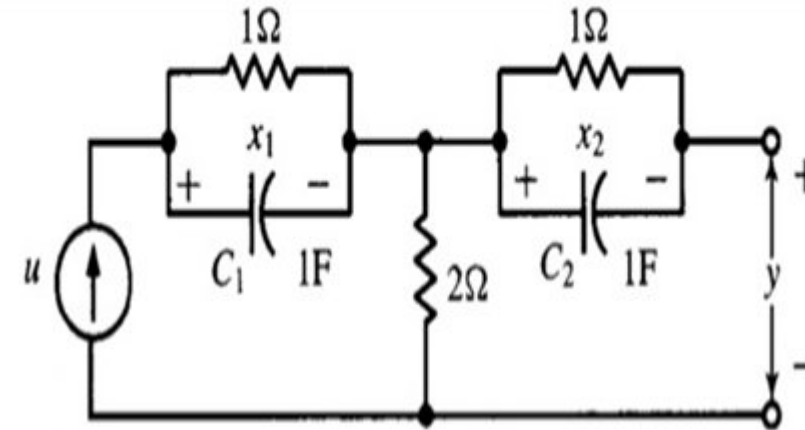
$$e^{At} = I \cos(\sqrt{2}t) + \frac{A}{\sqrt{2}} \sin(\sqrt{2}t)$$

STATE VARIABLE MODELS

Home-work



Find a state space model for the following electrical network. Let the output of the system be $y(t)$.





THANK YOU

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